Tribhuvan University

An Internship Report

As

"Python Developer"

At

Brand Builder Nepal

Submitted to

Institute of Science and Technology

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In partial fulfillment of the requirement for the Bachelor Degree in Computer Science and Information Technology

Submitted by

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DECLARATION

I hereby declare that the internship report on a project entitled **Resume AI** submitted to the Faculty of Science and Technology, Tribhuvan University, Kathmandu is a collaborative effort undertaken during my internship at **Brand Builder Nepal**. It is an original piece of work under the guidance of **Mr. Pukar Khanal** and supervision of **Mr. Raghu Gyawali** and faculty members, SOCH COLLEGE OF IT, RANIPAUWA, POKHARA, and is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Science in Computer Science and Information Technology. This internship report has not been submitted to any other university or institution for the award of any degree or diploma.

Submitted By:

Milan Devkota (27632/077)

B.Sc.CSIT, 2077 Batch

Soch College of IT, Pokhara

MENTOR'S RECOMMENDATION

SUPERVISOR'S RECOMMENDATION

CERTIFICATE OF APPROVAL

ACKNOWLEDGEMENT

I wish to express my sincere and profound gratitude to the individuals and organizations whose guidance, support, and opportunities have been instrumental in the successful completion of my internship and this report.

First and foremost, I extend my deepest appreciation to Mr. Pukar Khanal, Software Engineer at Brand Builder Nepal. Their expert guidance, continuous encouragement, and invaluable mentorship provided a strong foundation for my learning and professional development. I am particularly grateful for their willingness to entrust me with meaningful responsibilities and for their insightful feedback, which has significantly enriched my internship experience.

I would also like to express my sincere gratitude to Mr. Aashish Bijukchhe , Chief Executive Officer of Brand Builder Nepal. I am thankful for their leadership in creating a company that values learning and provides students with such significant opportunities for practical experience and growth through its internship program.

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Sincerely,

Milan Devkota

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ABSTRACT

This internship report details a project undertaken at Brand Builder Nepal during the internship period of 11 weeks, focusing on the development and enhancement of the AI-driven job recruit platform. This project presents **Resume AI**, an AI-driven resume analysis and job-matching platform designed to streamline the recruitment process for both job seekers and employers. Leveraging advanced natural language processing (NLP) and machine learning algorithms, Resume AI automatically parses, analyzes, and scores resumes based on relevance to specific job descriptions. The platform offers intelligent keyword matching, skill gap identification, and personalized feedback to help candidates optimize their resumes for better visibility and alignment with employer expectations. On the employer side, it enables faster shortlisting by ranking candidates based on key role-specific criteria, significantly reducing manual screening efforts. The project integrates real-time resume evaluation, intuitive UI/UX design, and scalable backend architecture to ensure performance and accessibility. By automating and enhancing critical stages of the hiring workflow, Resume AI bridges the gap between talent and opportunity, making recruitment more efficient, data-driven, and inclusive.

Keywords: Artificial Intelligence (AI), Job Matching, Natural Language Processing (NLP), Machine Learning, Applicant Tracking System (ATS), Recruitment Automation, Intelligent Hiring, Resume Optimization.

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence

API – Application Programming Interface

CV – Curriculum Vitae

HTML – HyperText Markup Language

HTTP – HyperText Transfer Protocol

NLP – Natural Language Processing

PDF – Portable Document Format

URL – Uniform Resource Locator

GUI – Graphical User Interface

DOM – Document Object Model

OCR – Optical Character Recognition

IP – Internet Protocol

IDE – Integrated Development Environment

CSS – Cascading Style Sheets

JS – JavaScript

SQL – Structured Query Language

JSON – JavaScript Object Notation

DB – Database

ML – Machine Learning

OS – Operating System

CLI – Command Line Interface

UI – User Interface

UX – User Experience

VS Code – Visual Studio Code

PDFs – Portable Document Formats

1. INTRODUCTION

1.1. Introduction

An internship serves as a critical stepping stone between academic knowledge and professional application, offering students and recent graduates an opportunity to gain hands-on experience, develop industry-relevant skills, and understand real-world workflows. It not only enhances technical capabilities but also fosters soft skills such as communication, problem-solving, and teamwork, which are essential for career success. (Adnan Khalid Jawabri, 2017)

In the dynamic landscape of recruitment technology, artificial intelligence is rapidly transforming how organizations identify, evaluate, and hire talent. To keep pace with this transformation, the internship at Brand Builder Nepal focused on the design and development of Resume AI, an intelligent, AI-powered platform aimed at streamlining the resume screening and job-matching process.

Resume AI is a cutting-edge web-based application developed to automate and enhance candidate evaluation using advanced Natural Language Processing (NLP) and machine learning algorithms. The system enables employers to process and rank resumes based on relevance to specific job descriptions, while simultaneously offering real-time feedback and optimization tips to job seekers. This dual-function approach not only reduces the manual workload of recruiters but also empowers candidates to improve their resume quality and visibility.

The project, carried out under the guidance of Brand Builder Nepal, involved the full-stack development of the platform—ranging from backend architecture and API integration to frontend design and user experience optimization. Key components included resume parsing, keyword matching, scoring algorithms, and the integration of intelligent feedback loops. The system also ensures compatibility with modern Applicant Tracking Systems (ATS), further improving its practical applicability in real-world hiring environments.

This internship provided a valuable opportunity to engage in a technically challenging, industry-relevant project that aligns with current trends in HRTech and AI-driven recruitment solutions.

1.2. Problem statement

The internship, conducted over a limited duration of approximately 10 to 11 weeks, presented several challenges that impacted the overall learning experience. One of the primary difficulties was the time constraint, which limited the depth of exploration and implementation of more advanced concepts and tools. At the initial stage, there were noticeable knowledge gaps, particularly in the use of technologies such as Selenium, FastAPI, and automation frameworks, which required additional self-learning and adaptation. Operating in a remote or semi-remote environment also posed communication challenges, often leading to delays in feedback and reduced clarity in task requirements. Balancing internship responsibilities with ongoing academic commitments demanded effective time management and task prioritization. Furthermore, transitioning from an academic background to a professional setting revealed a lack of practical exposure to industry-standard practices, including modular coding, documentation, and collaborative development workflows. Technical issues, such as environment setup errors and debugging complexities, further added to the workload and required persistent problem-solving. Despite these obstacles, the internship experience served as a valuable opportunity to develop both technical skills and professional competencies through real-world challenges and continuous learning.

1.3. Objectives

The main objective of the internship at Brand Builder Nepal is to gain practical experience in a real-world tech environment, develop technical and professional skills in AI-driven automation, and build industry networks to enhance career prospects. The specific objectives of the internship project are:

- To develop a LinkedIn Job Application Bot using Selenium for automating job application processes, thereby enhancing programming proficiency.
- To integrate Google Gemini via LangChain for generating tailored responses to job application questions, advancing expertise in artificial intelligence applications.
- To construct a scalable FastAPI backend for the bot to ensure efficient execution and status monitoring, strengthening backend development capabilities.

1.4. Scope and Limitation

1.4.1 Scope

The internship at Brand Builder Nepal offered comprehensive, hands-on experiences in automation, artificial intelligence, AI Integration. The scope included:

- Gaining practical exposure to real-world software development workflows, including version control, debugging, and deployment practices.
- Enhancing technical proficiency through continuous learning of industry-relevant tools, libraries, and frameworks used in automation and AI development.
- Developing professional communication, collaboration, and time management skills by actively participating in team discussions, remote coordination, and task reporting.

1.4.2 Limitation

Despite the extensive learning opportunities, the internship also has some limitations:

- The limited duration of the internship (three months) restricted deeper exploration of advanced AI techniques.
- Platform constraints, particularly with LinkedIn’s anti-bot mechanisms, prevented full automation of the job application process.
- As an intern, the scope for making independent decisions was also limited, with major decisions often requiring approval from mentor or clients.

1.5. Report Organization

Chapter 1 introduces the internship program, providing a clear overview of the problem statement, objectives, scope, and limitations of the internship. This chapter also outlines the structure of the report and briefly describes the content that will be covered in the subsequent chapters.

Chapter 2 presents a detailed analysis of the organization, including its organizational hierarchy, motivation, services offered, and key departments. This chapter also includes a literature review relevant to the organization's field and the internship experience.

Chapter 3 focuses on the specific roles and responsibilities undertaken during the internship. It includes a weekly log of activities, a description of the projects involved, and the tasks performed. Additionally, this chapter provides an in-depth look at the project implementation, including the tools and platforms used.

Chapter 4 concludes the report by summarizing the entire internship experience. It highlights the key learning outcomes, the skills developed, and the insights gained from the internship, offering a reflection on the overall experience and its impact on personal and professional growth.

2. ORGANIZATION DETAILS AND LITERATURE REVIEW

2.1. Introduction to Organization

Brand Builder Nepal is a well-known software and digital marketing privately owned company based in Pokhara, Nepal. They focus on creating brands, developing web and mobile applications for iOS and Android, and providing UI/UX design and digital marketing services. Their goal is to help businesses enhance their online presence and achieve better market visibility through innovative technology and effective marketing strategies.

The company is recognized for delivering high-quality services in software development, branding, and marketing. Brand Builder Nepal's office is located in Niva Galli, Chipledhunga, making them a central figure in Pokhara's growing digital landscape.

2.1.1. Organization Contact Details

Fig 2.1: Logo of Brand Builder Nepal Pvt. Ltd

Table 2.1: Organization Details

Name	Brand Builder Nepal Pvt. Ltd
Address	Niva Galli, Chipledhunga, Nepal
Phone	+977 9802850777
Email	mail@brandbuilder.com.np
Website	www.brandbuilder.com.np
Opening hours	10:00 am – 5:00 pm

2.1.2. Organization Mission, Vision and Objective

- **Mission:** Brand Builder aims to revolutionize brand creation by seamlessly integrating software technologies with the expertise of developers, content creators, SEO planners, and graphic designers in order to innovate and empower businesses with dynamic branding solutions ensuring visibility, engagement, and growth.

- **Vision**: The Software Company's vision is to be the beacon for businesses seeking to 'Stand Out of the Crowd' by blending cutting-edge software technologies.

- **Objective:** The primary objective of the company is to provide quality IT solutions and services to clients within Nepal and abroad. However some of the other objectives are as follows:
  - To develop innovative software, applications, and technology to solve business challenges.
  - To create a skilled workforce in the IT industry in Nepal and provide employment opportunities for Nepalese youth.
  - To establish a strong brand presence in the domestic and international markets and gain recognition as a reliable IT service provider.
  - To engage customers with the brand through various digital channels such as social media, email marketing, and content marketing. (Brand Builder Nepal, 2024)

2.1.3 Motivation for Choosing Brand Builder Nepal

Brand Builder Nepal was selected as the internship organization due to its strong alignment with core interests in Python programming, automation, and artificial intelligence. The company's focus on developing innovative technology solutions, particularly products like Resume AI, which emphasize real-world AI applications offered a unique opportunity to gain hands-on experience in a dynamic environment.

The chance to contribute to projects involving automation, web development, and AI integration aligned well with the goal of enhancing technical and problem-solving competencies. Furthermore, Brand Builder Nepal's reputation as a leading technology company in Pokhara, supported by a team of skilled professionals, made it an ideal setting for practical learning and professional growth.

Special appreciation is extended to Mr. Bijukchhe, CEO of Brand Builder Nepal, for providing this opportunity and fostering an environment where innovation and continuous learning are encouraged.

2.1.4 Organization Hierarchy

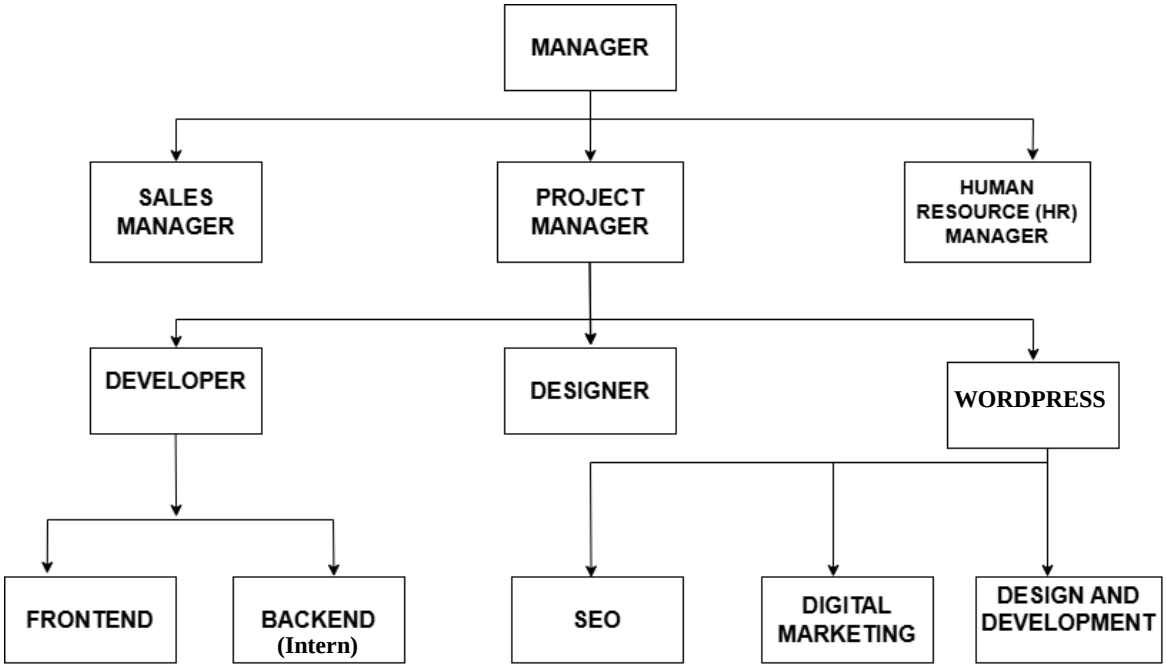

Fig 2.2: Brand Builder Nepal Organization Hierarchy

The organizational structure of Brand Builder Nepal features a clear hierarchy, starting with a central Manager who oversees several key departments. Reporting directly to the Manager are the Sales Manager, Project Manager, and Human Resource (HR) Manager, each responsible for their respective areas. The Project Manager then supervises the Developer team, which is specialized into Frontend and Backend functions, as well as the Designer team. Furthermore, the Project Manager also manages the WordPress team, which integrates both Design and Development aspects to deliver comprehensive web solutions. This structure highlights a focused approach to managing different aspects of the business, from sales and project execution to technical development and website management.

2.1.5 Working Domains of Organization

Brand Builder Nepal is dedicated to helping businesses enhance their online presence through a wide range of services, including web and mobile app development, WordPress

solutions, SEO, digital marketing, graphic design, and UI/UX design. These core domains reflect Brand Builder Nepal's commitment to delivering comprehensive and innovative digital solutions to its clients.

2.1.6 Description of Intern Department/Unit

During the internship at Brand Builder Nepal, the intern was an integral part of the Developer team, which was divided into Frontend and Backend specializations. Within the Backend division, the focus was on Python development, with a primary emphasis on creating automation solutions. Key responsibilities included automating the LinkedIn job application process, developing an intelligent system for form completion and resume parsing, integrating a Language Model for autonomous response generation, and ensuring the system's robustness through error handling and logging. These tasks contributed significantly to streamlining workflows and improving operational efficiency within the company.

2.1.7 Internship Durations

The internship began on 23rd February 2024 and lasted for 11 weeks, with 6 working hours per day, 6 days a week. It was conducted under the mentorship of Mr. Pukar Khanal, who provided valuable guidance throughout the program.

Table 2.1: Internship Durations

Start Date	23 Feb, 2024
Internship Duration	11 Weeks
Office Hours	11:00 am - 5: 00 pm
Working Hours	6 hours per day
Working Days	6 days a week
Mentor	Mr. Pukar Khanal

2.2. Literature review

Automation is reshaping human resources (HR), particularly in recruitment, by enhancing efficiency and streamlining candidate management. Smith and Brown (2021) found that automated tools, such as applicant screening software, reduce hiring time by 30%, though they may impersonalize candidate interactions. Applicant Tracking Systems (ATS) are central to this process, yet Lee and Patel (2020) note their keyword-based filters often reject qualified candidates, with a 20% false-negative rate. Python-based tools, like Selenium and Pandas, address these limitations by automating data extraction and form filling, cutting task time by 40% (Patel, 2022). This aligns with my internship work, where I explored automation solutions to optimize recruitment processes

Natural Language Processing (NLP) and Language Models (LLMs) further enhance recruitment by parsing resumes and generating application responses. Garcia and Lee (2022) report that transformer-based NLP models achieve 85% accuracy in extracting resume data, though biases in training data pose challenges. Wang et al. (2023) demonstrate LLMs' effectiveness in automated form filling, with a 90% success rate, provided prompts are carefully designed. Web automation tools like Selenium facilitate these tasks but raise ethical concerns, such as data privacy and compliance with platform terms (Brown & Taylor, 2023). These technologies informed my internship project, particularly in developing efficient application workflows.

Ethical considerations are critical in AI-driven recruitment. Miller and Kim (2022) found that 25% of AI hiring tools exhibit bias, necessitating robust ethical frameworks to ensure fairness and GDPR compliance. Research also highlights job seekers' frustration with repetitive online applications, with 70% reporting challenges on platforms like LinkedIn (Taylor & Singh, 2021). Automation can reduce application time by 50% (Davis & Lee, 2023), but over-automation risks depersonalization. These findings underscore the need for balanced automation, guiding my internship efforts to create ethical and efficient HR solutions.

3. INTERNSHIP ACTIVITIES

3.1 Roles and Responsibilities

During the internship, the primary focus was on automating the job application process on LinkedIn to improve efficiency and reduce manual effort. The core responsibility involved developing a comprehensive system to automate the entire workflow, from account login to targeted job searching and submitting “Easy Apply” applications. This was achieved by utilizing advanced web scraping techniques to extract relevant job data and automate navigation through LinkedIn’s interface, ensuring a seamless and efficient application process.

Additionally, an intelligent system was developed to automate the completion of application forms by parsing key data from resumes to accurately populate standard fields and manage file uploads. This reduced manual data entry and improved the overall speed and accuracy of the process. To further enhance the system’s capability, a Language Model was integrated to generate tailored responses for open-ended application questions, using data extracted from resumes. This integration enabled the system to handle complex application requirements with greater precision.

To ensure the system’s reliability and smooth operation, robust error handling, detailed logging, and strategic delays were implemented. These measures minimized disruptions, optimized performance, and maintained discretion during operations. The resulting automation tool significantly enhanced the job application workflow, providing a streamlined, efficient, and user-friendly solution for applying to jobs on LinkedIn.

3.2 Weekly Logs

Table 3.1: First week log

Days	Activities
Day 1	Commenced internship with introduction and project overview. Set up development environment.
Day 2	Conducted initial research on LinkedIn's website structure to identify key HTML elements for job searching and application automation.

Day 3	Studied Selenium's documentation and tutorials to gain a comprehensive understanding of web automation techniques using Python.
Day 4	Planned the project's architecture, defining the necessary modules for login, search, filtering, application processing, GUI, and backend API communication.
Day 5	Established the project's repository using Git and created the initial code structure for the bot, GUI (HTML, CSS, JS), and backend (FastAPI).
Day 6	Configured environment variables to securely handle sensitive information and set up Python virtual environments to manage project dependencies.

Table 3.2: Second week log

Days	**Activities**
Day 1	Developed a function using Selenium to automate the login process on LinkedIn, including handling username and password input.
Day 2	Implemented cookie-based login functionality to persist user sessions and avoid repeated logins, enhancing efficiency.
Day 3	Set up the basic structure for the GUI using HTML, focusing on the layout and initial elements.
Day 4	Designed the GUI form to capture user input, including credentials, job search criteria, and resume data.
Day 5	Established a basic FastAPI application and defined the initial API endpoints required for communication with the GUI.
Day 6	Connected the GUI's login form to the FastAPI backend, enabling data transfer between the frontend and backend (without Selenium integration at this stage).

Table 3.3: Third week log

Days	Activities
Day 1	Developed the function to navigate to the LinkedIn Jobs section and initiate job searches using Selenium.
Day 2	Implemented the job search functionality, enabling users to specify job titles and locations through keywords.
Day 3	Added input fields to the GUI to allow users to enter their desired job title and location for the search.
Day 4	Connected the GUI's job search inputs to the FastAPI backend, which triggered the Selenium job search process.
Day 5	Implemented the display of job search results (initially job titles) within the GUI.
Day 6	Implemented error handling for the job search process (Selenium and API) and displayed error messages in the GUI.

Table 3.4: Fourth week log

Days	Activities
Day 1	Developed the function to filter job listings for "Easy Apply" jobs.
Day 2	Integrated the "Easy Apply" filter option into the GUI.
Day 3	Implemented file upload functionality in the GUI to allow users to upload their resume files.
Day 4	Configured the FastAPI backend to receive and temporarily store the uploaded resume file.
Day 5	Designed the data structure (JSON format) to store and manage resume information within the backend.
Day 6	Implemented the function to load resume data from the uploaded file and store it in the backend.

Table 3.5: Fifth week log

Days	Activities
Day 1	Developed the core logic to iterate through job listings and initiate the application process using Selenium.
Day 2	Analyzed LinkedIn's Easy Apply forms to identify common form fields and their HTML elements.
Day 3	Implemented functions to automatically fill basic application form fields using data from the user's resume.
Day 4	Created API endpoints in FastAPI to trigger the application process for a selected job listing.
Day 5	Connected the GUI to the FastAPI application endpoints to initiate job applications.
Day 6	Conducted initial testing of the application automation for basic fields using sample Easy Apply job postings.

Table 3.6: Sixth week log

Days	Activities
Day 1	Implemented functionality to handle resume uploads within the application process using Selenium.
Day 2	Addressed scenarios with conditional form fields or variations in application structures, implementing initial handling strategies.
Day 3	Developed a mechanism within FastAPI to track the status of each application and log attempts.
Day 4	Incorporated delays and waiting times into the bot's actions to mimic human behavior.
Day 5	Refined the GUI by adding progress indicators and detailed status messages.
Day 6	Improved the CSS styling of the GUI for better user experience.

Table 3.7: Seventh week log

Days	Activities
Day 1	Integrated LangChain with the FastAPI backend to handle custom application questions.
Day 2	Set up the LLM (Gemini) and defined prompts to generate answers based on resume data.
Day 3	Implemented logic to identify and extract custom questions from application forms using Selenium and FastAPI.
Day 4	Developed the function to generate answers using LangChain and automatically fill them into the form.
Day 5	Began exploring basic AI/ML concepts: supervised vs. unsupervised learning.
Day 6	Gained a basic understanding of Natural Language Processing (NLP): text processing, tokenization.

Table 3.8: Eighth week log

Days	Activities
Day 1	Implemented robust error handling to manage exceptions during the application process.
Day 2	Added comprehensive logging to record detailed information about the bot's actions and errors.
Day 3	Refined the bot's behavior to handle edge cases and unexpected scenarios.
Day 4	Improved the bot's overall efficiency and speed by optimizing code and minimizing waits.
Day 5	Enhanced the bot's ability to adapt to potential changes in LinkedIn's user interface.
Day 6	Conducted thorough testing of the entire system (GUI, API, Selenium automation).

Table 3.9: Ninth week log

Days	Activities
Day 1	Implemented advanced GUI features: progress bars, detailed logs, user settings.
Day 2	Added basic user authentication to the GUI and FastAPI backend.
Day 3	Optimized the GUI for responsiveness across various screen sizes.
Day 4	Prepared the FastAPI application for deployment, exploring Docker.
Day 5	Documented the API endpoints and GUI functionality.
Day 6	Created a detailed deployment plan and checklist.

Table 3.10: Tenth week log

Days	Activities
Day 1	Implemented enhanced handling for conditional logic in application forms.
Day 2	Developed a module to extract and save application confirmation details after submission.
Day 3	Integrated a mechanism to handle applications requiring external website redirects.
Day 4	Optimized the bot's performance by implementing asynchronous task execution for certain operations.
Day 5	Improved the bot's resilience to website changes by using more robust element selection strategies.
Day 6	Conducted focused testing on the application handling module to identify and resolve any edge cases.

Table 3.11: Eleventh week log

Days	Activities
Day 1	Enhanced the GUI to display detailed application status updates to the user.
Day 2	Implemented user settings within the GUI to allow customization of bot behavior.
Day 3	Refactored the FastAPI backend to improve code organization and maintainability.
Day 4	Integrated a rate-limiting mechanism into the backend to prevent excessive requests to LinkedIn.
Day 5	Implemented logging within the GUI to provide users with insights into the bot's actions.
Day 6	Conducted integration testing of the GUI and backend to ensure seamless communication and data flow.

3.3 Description of the Project Involved During Internship

During the internship, significant contributions were made to the development of an advanced automation system aimed at streamlining and enhancing the efficiency of the job application process on the LinkedIn platform. The system was engineered to automate key tasks including user account login, execution of targeted job searches based on specified criteria, and submission of applications for positions featuring LinkedIn's "Easy Apply" functionality.

A major component of the system involved the development of an intelligent form completion module. This module was designed to extract relevant data from the user's resume, accurately populate standard application fields, and manage the upload of required documents, substantially reducing the need for manual input.

Additionally, a Language Model (LLM) was integrated to generate contextually appropriate responses to open-ended application questions. These responses were crafted using information intelligently parsed from the user's resume, thereby increasing the personalization and relevance of each application.

To ensure the system operated reliably and discreetly, robust error-handling routines were implemented, along with detailed diagnostic logging and strategic time delays. These features collectively enhanced system stability and minimized the risk of detection or disruption during automated operations.

3.3.1. Introduction to Projects Involved

Project 1: Resume AI

Resume AI is an online platform designed to assist individuals in creating professional and effective curriculum vitae (CVs) or resumes with the power of Artificial Intelligence. The core aim of the service is to streamline and enhance the resume-building process, making it quicker and easier for users to produce compelling application documents. By leveraging AI, Resume AI helps users craft content, select appropriate templates, and optimize their resumes for modern job application standards.

The platform typically offers several key features centered around its AI capabilities. This includes an AI writer that can help generate or refine sections like professional summaries, experience bullet points, and skill descriptions, often tailored to specific job roles or industries. Resume AI also emphasizes ATS (Applicant Tracking System) compatibility, meaning its templates and AI suggestions are geared towards helping resumes pass through automated screening systems used by many employers. Users can usually choose from a variety of professional templates, customize layouts, and see real-time previews of their document.

Ultimately, Resume AI is targeted at job seekers who want to create a polished and impactful resume efficiently. Whether it's students crafting their first CV, professionals looking to switch careers, or anyone needing to update their existing resume, the tool aims to save time and reduce the stress associated with resume writing. By providing AI-driven suggestions and ATS-friendly designs, it strives to increase the user's chances of getting noticed by recruiters and securing interviews. Many such platforms offer a freemium model, allowing basic use for free with premium features available through paid subscriptions.

3.3.1.1 Problem statement

In the contemporary and highly competitive job market, the effective presentation of one's qualifications through a curriculum vitae (CV) or resume is paramount. However, a significant challenge for many job seekers lies in their ability to craft documents that are not only professionally formatted and grammatically impeccable but also strategically optimized to capture the attention of recruiters and navigate Applicant Tracking Systems (ATS). Individuals often lack the specialized expertise, time, or up-to-date knowledge of current best practices in resume writing, including keyword optimization and impact-driven content creation. This deficiency can lead to the submission of suboptimal application materials, thereby hindering their ability to effectively articulate their value proposition and secure interview opportunities, despite possessing relevant skills and experience.

3.3.1.2. Objectives

The main objectives of internship project were:

- To automate the LinkedIn job application process, enhancing efficiency from account login and job search to submitting "Easy Apply" applications.
- To develop an intelligent system for auto-completing application forms, parsing resume data to populate fields and manage file uploads, reducing manual input.
- To integrate a Language Model for generating tailored responses to open-ended application questions, leveraging resume data.
- To ensure system robustness and efficiency through error handling, logging, and strategic delays, minimizing disruptions and ensuring smooth operation.

3.3.1.3 Scope and Limitation

Scope

The scope of the project includes the following objectives:

- Empowering job seekers with AI-driven resume creation by providing an intelligent platform that simplifies and enhances the resume crafting process.
- Improving resume quality and impact through AI-powered content suggestions, optimization techniques, and access to professional templates.

- Increasing ATS compatibility by optimizing resumes for Applicant Tracking Systems, thus improving the chances of passing initial screenings.
- Reducing time and effort in job application preparation by streamlining the process, enabling efficient tailoring of resumes for multiple job applications.

Limitation

Despite the extensive scope, several limitations affect the project's execution. The limitations are described below:

- The quality of AI-generated resumes is heavily dependent on the accuracy and completeness of user input, with vague information potentially leading to generic outcomes.
- AI may lack the contextual understanding necessary to capture a user's unique career journey or personal brand, particularly for specialized roles requiring human insight.
- Over-reliance on AI can result in resumes that, while well-formatted, may lack individuality and fail to capture the attention of human reviewers.

3.3.1.4 Requirement Analysis

Functional Requirements

- Users should be able to upload CVs in common formats such as PDF and DOCX, which the system will automatically process and store securely.
- The system should parse uploaded CVs using Natural Language Processing (NLP) to extract relevant information such as name, contact details, skills, education, and work experience.
- Recruiters should be able to search and filter CVs based on keywords, required skills, years of experience, education level, and other custom criteria relevant to job roles.
- Each CV should be scored by an AI model based on how well it matches the criteria of a specific job posting, helping recruiters quickly identify the most suitable candidates.

- Users should be able to create and manage job postings, specifying job titles, descriptions, required qualifications, and skill sets, to which CVs can be matched and grouped.
- The system should allow users to view detailed profiles of applicants, including the parsed information, AI scores, and attached CV files.
- Admins should be able to manage user accounts and roles, monitor system activity, and oversee the overall operation of the platform to ensure quality and security.
- The platform should allow exporting of applicant data and CV analytics in formats like CSV or Excel for reporting, offline review, or integration with external HR systems.

Non-Functional Requirements:

- Scalability
  The system must efficiently handle increasing volumes of CVs, job listings, and user traffic, ensuring performance is not compromised as the platform scales.
- Security
  User data and CVs must be encrypted both at rest and in transit, following best practices to maintain privacy, comply with regulations like GDPR, and prevent unauthorized access.
- Usability
  The platform should offer an intuitive, user-friendly interface across all devices, with clear error messages, helpful documentation, and easy navigation to enhance the user experience.
- Maintainability
  The system must have a modular architecture, allowing for easy updates, enhancements, and debugging. The codebase should be well-documented for efficient future maintenance and scalability.
- Availability
  The system should ensure a minimum uptime of 99.9%, excluding scheduled maintenance, to support continuous job application processes without interruption.

- Reliability

  The AI model and resume parsing algorithms must consistently produce accurate and stable results, ensuring high-quality data processing for a variety of CV formats and job listings.

- Compatibility

  The platform must be fully compatible with major modern browsers (Chrome, Firefox, Edge, Safari), ensuring a seamless user experience across different devices and browsers.

By meeting these functional and non-functional requirements, Resume AI platform will provide a secure, efficient, and user-friendly experience for its customers.

Use Case Diagram

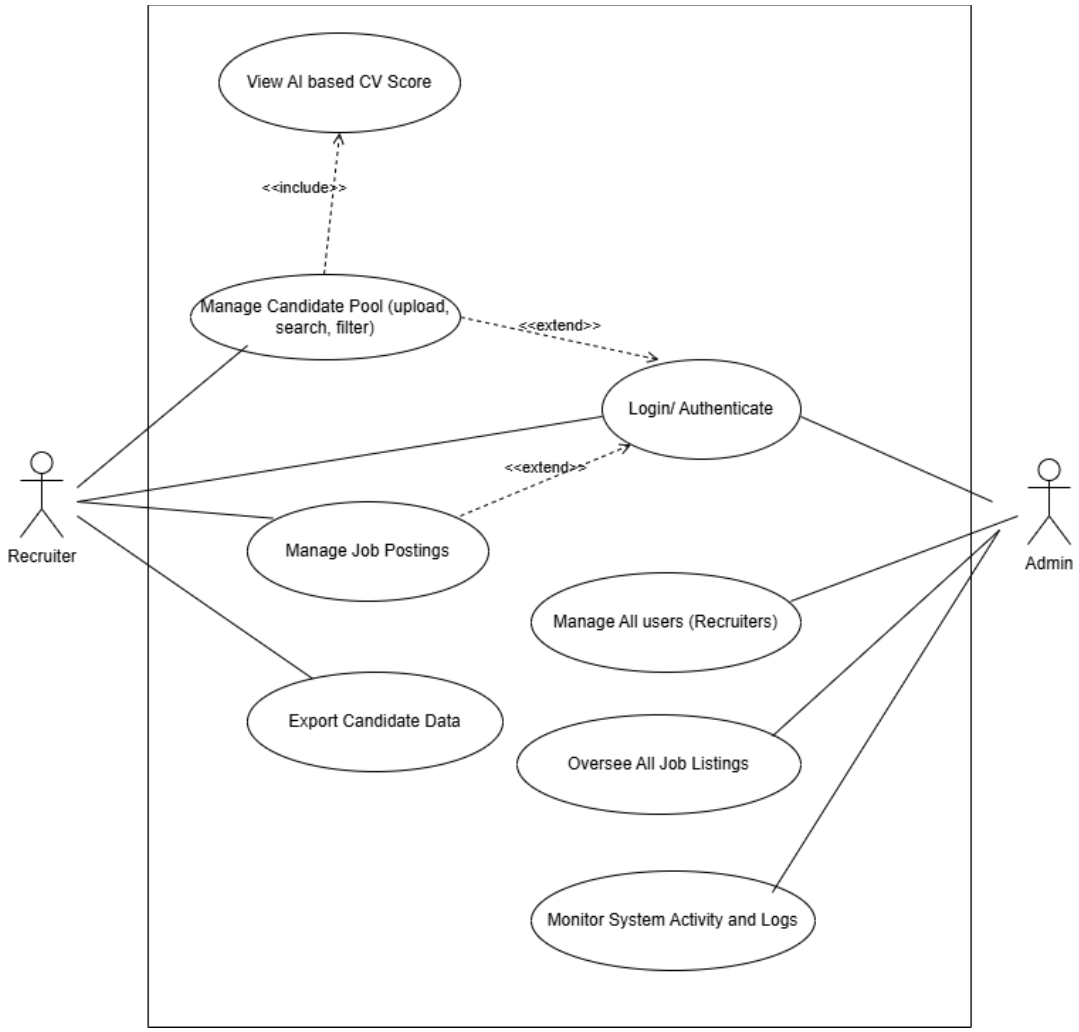

Fig 3.1: Use Case Diagram

This image presents a Use Case Diagram, though it lacks the critical actors and their connections to the use cases, which limits a complete interpretation. Nevertheless, the diagram outlines various functionalities of a system, likely related to recruitment or talent management. These include the ability to view an AI-based CV score, manage a pool of candidates (uploading, searching, filtering), handle user logins, create and manage job postings, administer all user accounts (specifically recruiters), export candidate data, oversee all job listings, and monitor system activity and logs. In essence, it shows a range of features enabling the system to support candidate processing, job management, and administrative oversight.

Activity Diagram for User

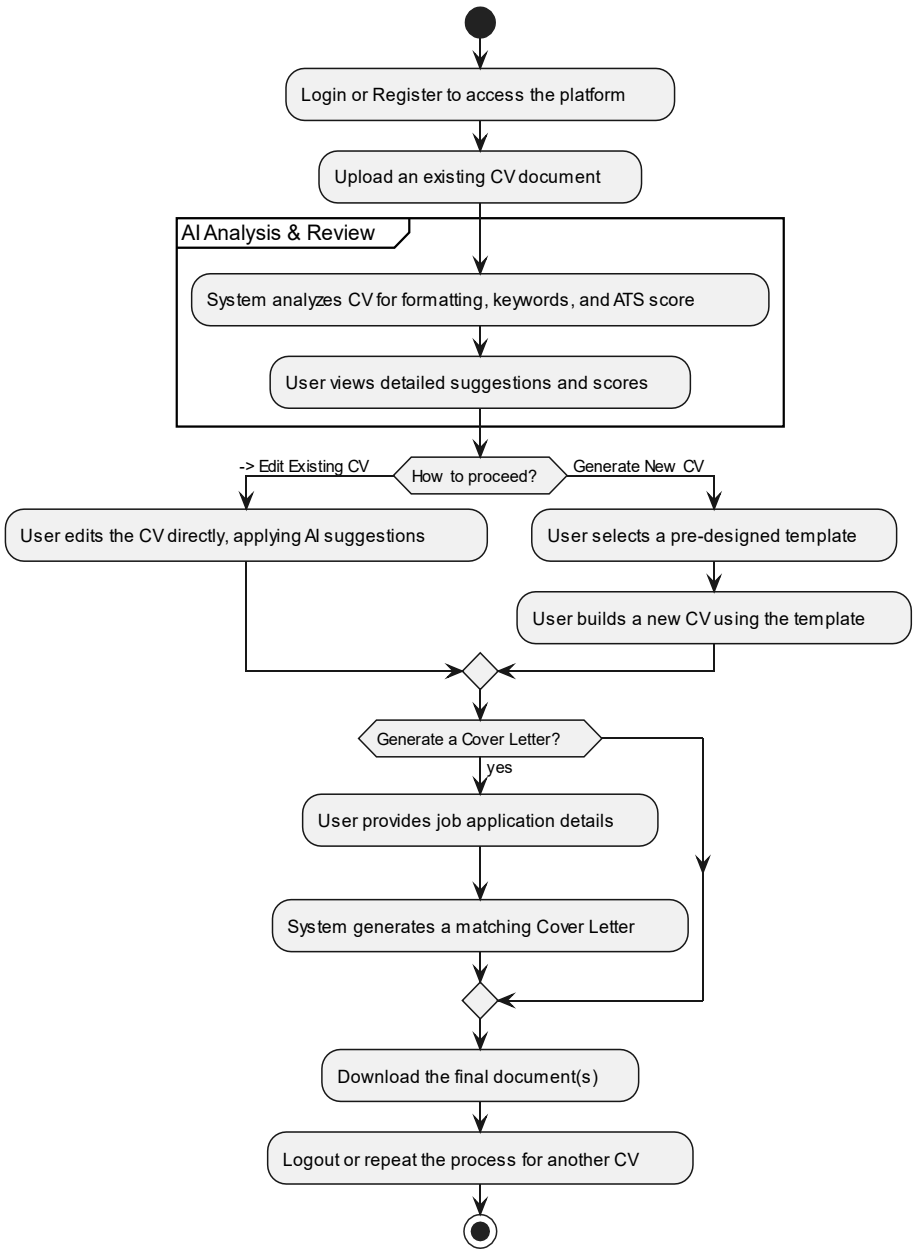

Fig 3.2: Activity Diagram for User

This activity diagram illustrates a user's journey through a platform designed to enhance CVs and create cover letters. Users begin by logging in and uploading their CV, which the system then analyzes for various criteria, providing detailed suggestions and an ATS score. Following this, users can choose to edit their CV directly with AI assistance, select a pre-designed template, or build a new CV using a template. Optionally, they can generate a

matching cover letter by providing job details. The process concludes with the user downloading their optimized document(s) and the option to log out or repeat the procedure for another CV.

Activity Diagram for Admin

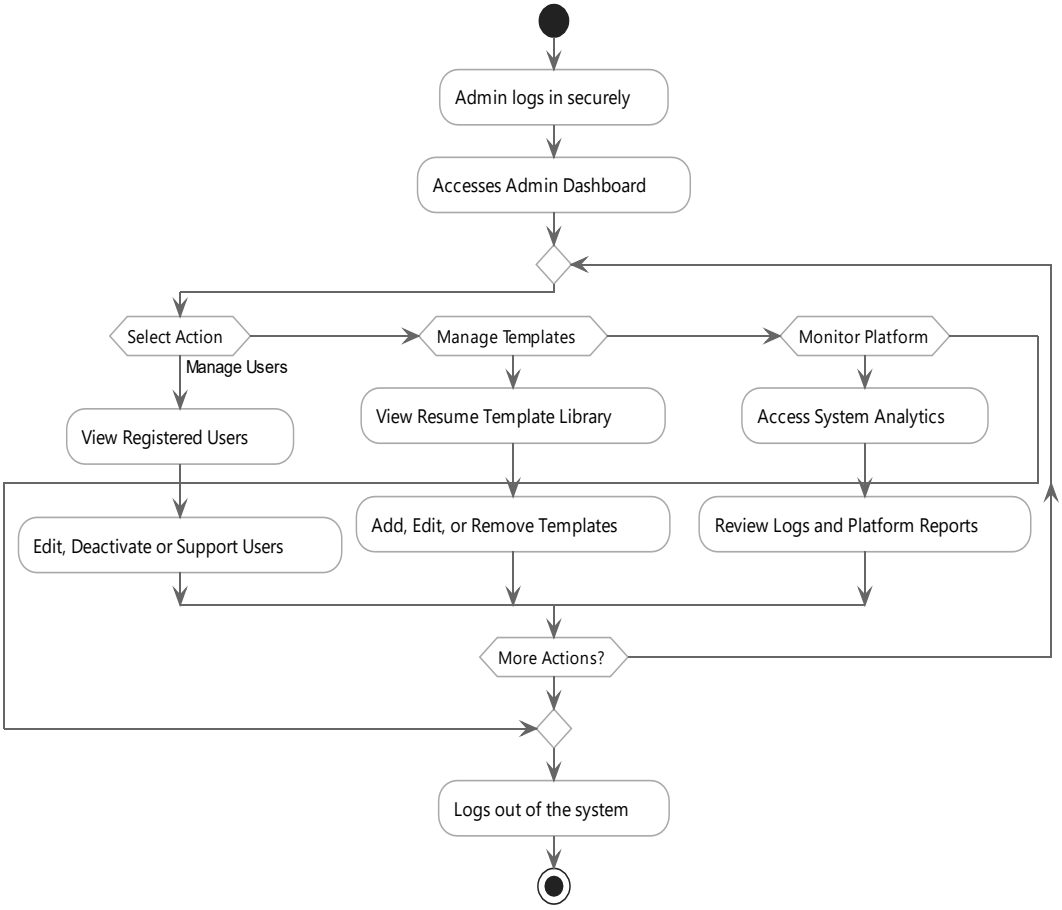

Fig 3.3: Activity Diagram for Admin

This activity diagram depicts the administrative workflow within a system. An administrator securely logs in to access a dashboard, from which they can manage registered users (viewing, editing, deactivating, supporting), oversee resume templates (viewing, adding, editing, removing), or monitor the platform's health (accessing analytics, reviewing logs and reports). After completing any of these actions, the admin decides whether to perform more tasks or log out of the system.

3.3.2. Implementation

The successful implementation of the Resume AI LinkedIn automation platform required the integration of several modern tools and technologies. These tools contributed to streamlining the job application process, enhancing automation, improving user interaction, and enabling AI-driven functionality. Below are the key technologies and their roles in the project:

Python and Selenium for Automation

Python was chosen as the primary programming language due to its simplicity and powerful ecosystem for automation. Selenium WebDriver was utilized to control the browser and perform automated tasks on LinkedIn, such as logging in, searching for jobs, and submitting applications. Selenium's capability to mimic user behavior (e.g., mouse movement, delay between actions) played a crucial role in reducing detection by LinkedIn's anti-bot systems.

FastAPI for Backend Development

FastAPI was employed to build a lightweight and high-performance backend API. It facilitated the connection between the frontend HTML form and the backend automation script. User input (e.g., LinkedIn credentials, job title, location) was securely passed from the interface to the backend, triggering the automation process. FastAPI’s asynchronous capabilities ensured efficient handling of multiple tasks and requests.

HTML/CSS for User Interface

An HTML form styled with CSS was developed to allow users to input necessary data such as LinkedIn login credentials, job titles, and job locations. This user-friendly interface made it easier to initiate automation without using the command line, improving accessibility for non-technical users.

LangChain and Gemini for AI-Generated Responses

To automate form filling during the job application process, LangChain was integrated with Google’s Gemini model. This AI-powered solution generated personalized and

contextually relevant responses to common application questions, enhancing the effectiveness and professionalism of each application submission.

Logging and Error Handling

A comprehensive logging system was implemented to track each step of the automation process, including job searches, application statuses, and any errors encountered. This made debugging easier and provided transparency into the execution flow.

Notion and Slack for Communication with Client

Effective collaboration with mentors and documentation of daily progress during the Resume AI internship was achieved using Notion and Slack:

Notion: Served as the central platform for documenting daily tasks, recording standup notes, and maintaining weekly progress logs. It enabled structured tracking of internship activities, learnings, and pending tasks in a collaborative and organized workspace.

Slack: Used for real-time communication with mentors and team members. It facilitated daily standups, quick feedback exchanges, and prompt resolution of blockers, ensuring consistent guidance and project alignment throughout the internship period.

3.3.3. Development Methodology

For the Resume AI automation platform project, we adopted the Agile development methodology using the Scrum framework. This approach allowed us to iterate quickly, adapt to evolving requirements, and receive continuous feedback from mentors, ensuring an optimized learning and development process.

- Weekly Standups and Review Sessions: At the end of each week, we conducted virtual review sessions with mentors. These meetings served as our Sprint Reviews and Sprint Planning sessions. During these meetings, we:
    - Reviewed Completed Tasks: Demonstrated working features such as login automation, job search, and auto-apply functionality.
    - Received Feedback: Mentors provided feedback on implementation, coding practices, and project structure.

    - Planned Next Sprint Goals: Tasks were assigned for the upcoming week, focusing on either refining current modules or developing new features.

- Daily Communication: Daily progress and blockers were shared via Slack, ensuring real-time support and guidance. This allowed prompt troubleshooting and uninterrupted progress.
- Sprint Execution: Each development sprint typically spanned 5–6 days. Tasks were selected from a shared Notion board and executed iteratively, ensuring that work was structured and goal-oriented.
- Adaptability: The Agile process allowed us to accommodate new features like AI integration (LangChain + Gemini) and frontend form support mid-way through the project, without disrupting progress.

By using the Agile development methodology with the Scrum framework, we ensured that the Resuma AI project remained flexible, client-focused, and efficiently managed. This methodology facilitated continuous improvement, regular feedback, and successful project outcomes.

3.4 Tasks / Activities Performed

During the internship, I was primarily involved in the development of an AI-powered LinkedIn job application automation platform. The project combined web automation, backend development, AI integration, and resume parsing functionalities. I also worked on documentation, daily reporting, and collaboration with the team using modern project management tools.

- **LinkedIn Development:** Designed and implemented a Python-based automation script using Selenium to log in to LinkedIn, search for jobs by title and location, and apply via the Easy Apply feature. Human-like interactions were simulated using randomized delays and intelligent element detection.
- **Backend Development with FastAPI:** Created a FastAPI-based backend service to connect the automation logic with a frontend interface. This allowed users to input credentials and job preferences securely, triggering the backend processes accordingly.

- **Frontend Integration:**Developed a simple HTML-based frontend for user input, enabling access for non-technical users. The system was tested across browsers for compatibility and responsiveness.
- **AI Response Generation:** Integrated LangChain and Google Gemini to dynamically generate personalized responses to application questions, reducing manual effort and improving the quality of applications.
- **Logging and Error Handling:** Implemented detailed logging for tracking key actions and errors. Added error-handling mechanisms to retry failed attempts and ensure smooth execution of the automation process.
- **Resume Parsing with NLP:** Partially developed a system using spaCy and PyPDF2 to extract keywords from PDF resumes, categorize applicants by job title, and prevent duplicate entries.
- **Project Management with Notion:** Used Notion to document daily progress, manage tasks, and record feedback. This ensured structured reporting and transparency throughout the internship.
- **Team Communication via Slack:** Maintained regular communication with mentors and teammates through Slack. Conducted daily standups, discussed blockers, and shared updates in real-time for better collaboration.

4. CONCLUSION AND LEARNING OUTCOMES

4.1. Conclusion

In conclusion, the internship at Brand Builder provided a significant opportunity to contribute to a cutting-edge project utilizing automation and artificial intelligence to streamline the LinkedIn job application process, a critical and impactful area in today's employment landscape. The internship allowed for the application of theoretical knowledge to real-world problems, leading to a deeper understanding of how technologies such as Selenium, FastAPI, and Language Models can be integrated to solve practical challenges efficiently.

Throughout the internship, technical skills in Python, automation, backend development, and natural language processing (NLP) were enhanced, alongside exposure to vital software engineering practices, including Agile project management, sprint tracking, and collaborative development. The structured mentorship, daily standups, and consistent feedback sessions facilitated growth in teamwork, problem-solving, and the ability to deliver scalable solutions in a time-constrained environment. These experiences cultivated a methodical approach to problem-solving, critical thinking under constraints, and effective iteration based on user feedback.

Moreover, the internship emphasized the importance of developing software that is not only functional but also secure, user-friendly, and adaptable. The inclusion of features like error handling, logging, dynamic CV parsing, and AI-generated responses highlighted the need for designing scalable systems with user experience in mind. This comprehensive approach to development has inspired further interest in artificial intelligence and automation, with a focus on applications that improve digital accessibility and productivity.

Overall, the internship at Brand Builder has been a pivotal experience in both academic and professional development. It reinforced a commitment to pursuing advanced studies and a career in software engineering. The skills and insights gained during the internship will provide a solid foundation for future endeavors in the tech industry and academia.

4.2. Learning Outcomes

The internship at Brand Builder Nepal marked a significant turning point in both academic and professional development. Throughout the program, extensive hands-on experience was gained in automation, backend development, and AI integration, providing a deeper understanding of full-stack application development and intelligent system design. A key learning was the practical application of Python and Selenium to automate complex web workflows, particularly for LinkedIn's job application interface. This involved simulating human-like behavior, managing dynamic content, handling exceptions, and implementing error-logging mechanisms for reliable automation scripts.

Exposure to FastAPI was equally valuable, as it allowed for the development of a lightweight, efficient backend service that bridged the automation engine and the user interface. This experience enhanced understanding of RESTful APIs, request/response handling, and deploying backend logic in a production environment, while integrating this service with an HTML frontend form improved accessibility and user experience.

The integration of Large Language Models (LLMs), specifically through LangChain and Google's Gemini, provided the opportunity to learn how to dynamically generate personalized responses for job applications, advancing skills in prompt engineering, NLP workflows, and the practical application of AI in recruitment automation.

In summary, the internship provided a comprehensive learning experience, blending theoretical knowledge with hands-on implementation. It fostered growth in both technical skills—such as automation, backend development, and AI integration—and essential soft skills like communication, collaboration, and time management. These experiences have solidified foundational capabilities as a developer and prepared for future academic and professional endeavors in AI, automation, and software engineering.

REFERENCES

Brand Builder Nepal Pvt. Ltd. (20240 - *Company Overview, Mission, Vision, and Objectives.* Retrieved from: https://brandbuilder.com.np/

Brown, A., & Taylor, R. (2023). Web automation in recruitment: Opportunities and ethics. *Journal of Technology and Management*, 18(2), 45–60. Retrieved from https://jtm.org/articles/2023/web-automation

Chaffey, D., & Ellis-Chadwick, F. (2019). *Digital Marketing: Strategy, Implementation and Practice*. Pearson. Retrieved from: https://www.scirp.org/reference/referencespapers?referenceid=3413574

Davis, R., & Lee, J. (2023). Automation and job application efficiency. *HR Technology Journal*, 25(1), 33–47. Retrieved from https://hrtj.org/2023/efficiency

Doyle, C., McDowell, D., McGarry, K., & Peppard, J. (2017). *Responsive Web Design: Key Considerations for a Successful Implementation*. ACM Computing Surveys, 50(1), 1-18. Retrieved from: https://www.researchgate.net/publication/3210546789_ResponsiveWeb Design_Key Considerations_for_a_Successful_Implementation

Ellis-Chadwick, F., & Doherty, N. F. (2012). Web advertising: The role of e-mail marketing. *Journal of Business Research*, 65(6), 843-848. Retrieved from: https://www.sciencedirect.com/science/article/abs/pii/S0148296311000063

Huang, Z., & Benyoucef, M. (2013). From e-commerce to social commerce: A close look at design features. *Electronic Commerce Research and Applications*, 12(4), 246-259. Retrieved from: https://www.sciencedirect.com/science/article/abs/pii/S156742231200124X

Jawabri, A. K. J. K. (2017). *(PDF) exploration of internship experience and satisfaction leading to better career prospects among business students in UAE*. researchgate. Retrieved from:https://www.researchgate.net/publication/321052709_Exploration_of_Internship_Experience_and_Satisfaction_Leading_to_Better_Career_Prospects_among_Business_Students_in_UAE

Wang, J., Zhang, L., & Chen, M. (2023). Language models in recruitment automation. *Machine Learning Applications*, 20(3), 67–82. Retrieved from https://mla.org/2023/llm-recruitment

APPENDIX

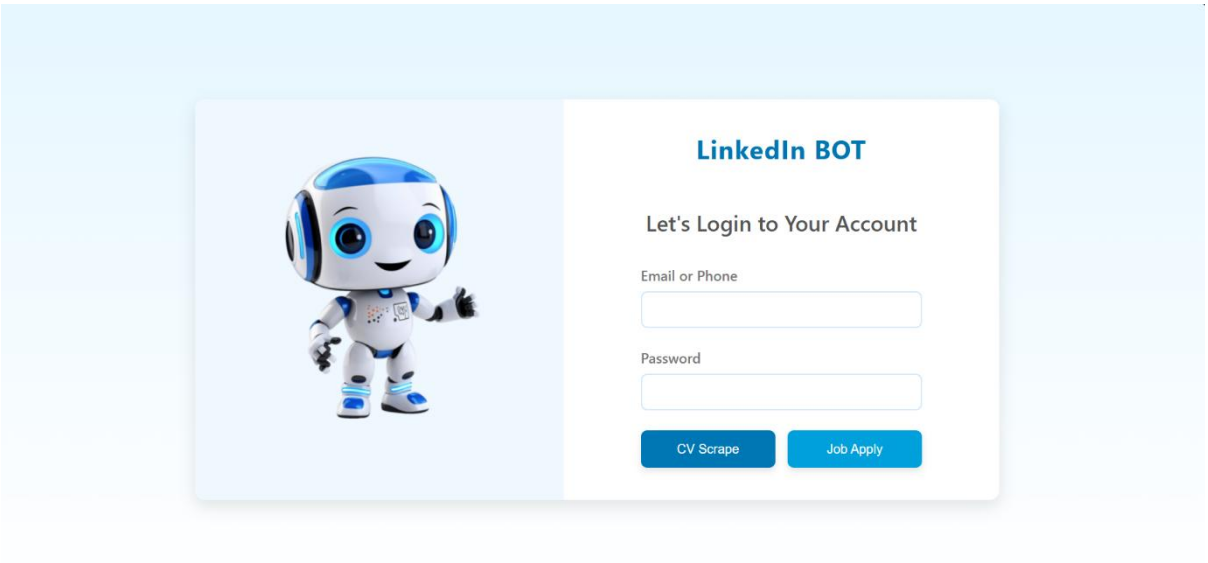

Fig: Login Page for LinkedIn bot

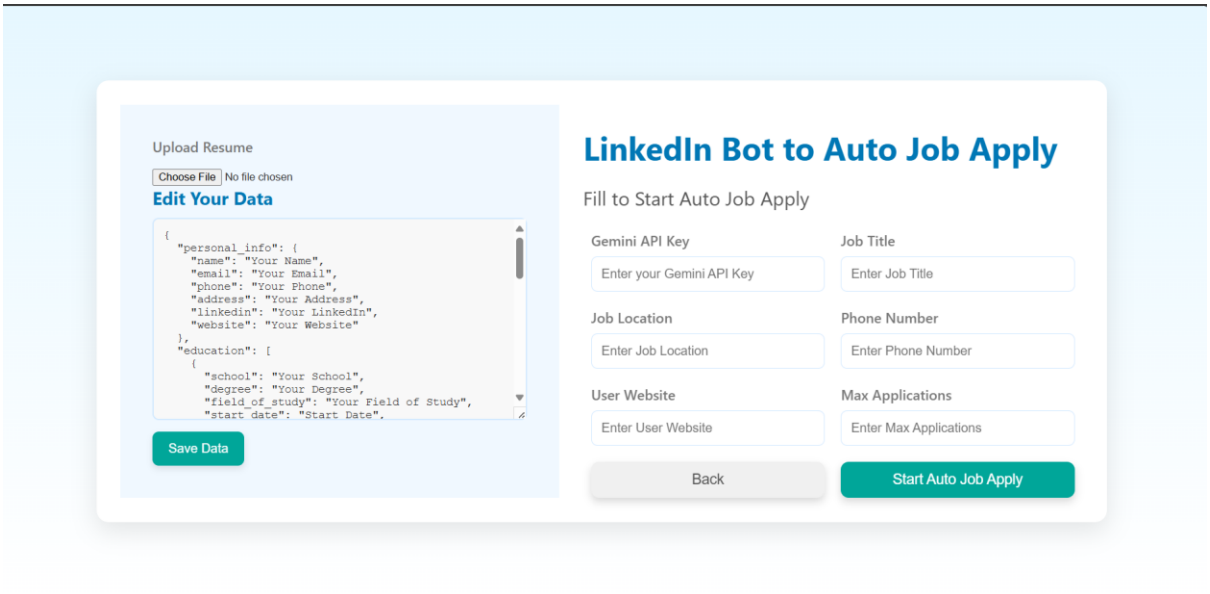

Fig: Form filing page for LinkedIn bot

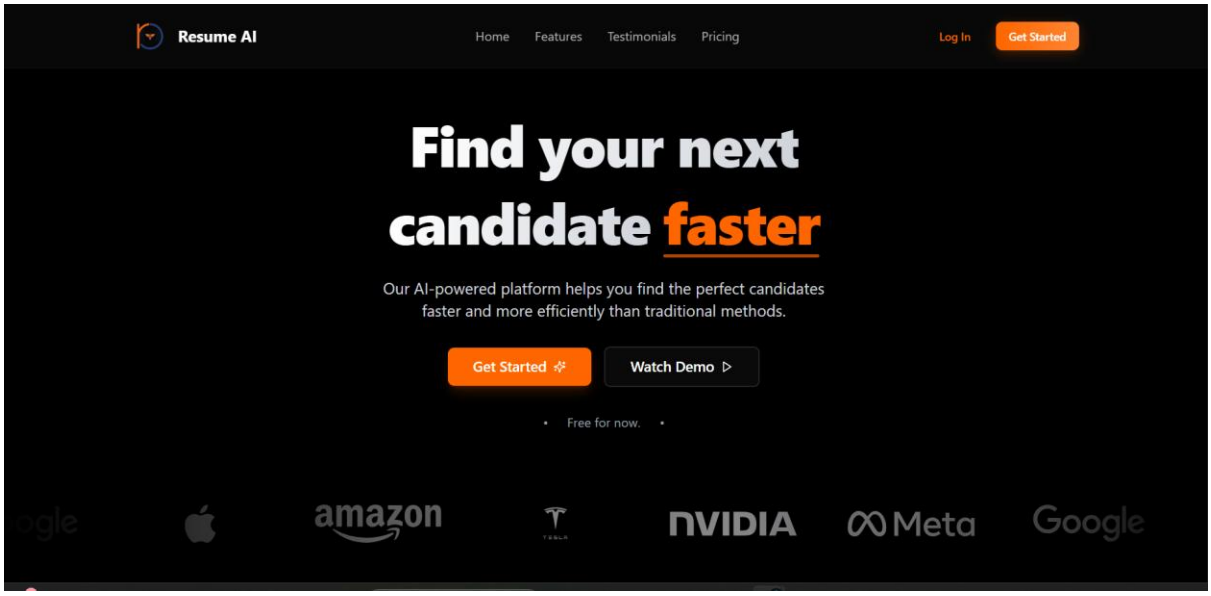

Fig: Resume AI landing Page

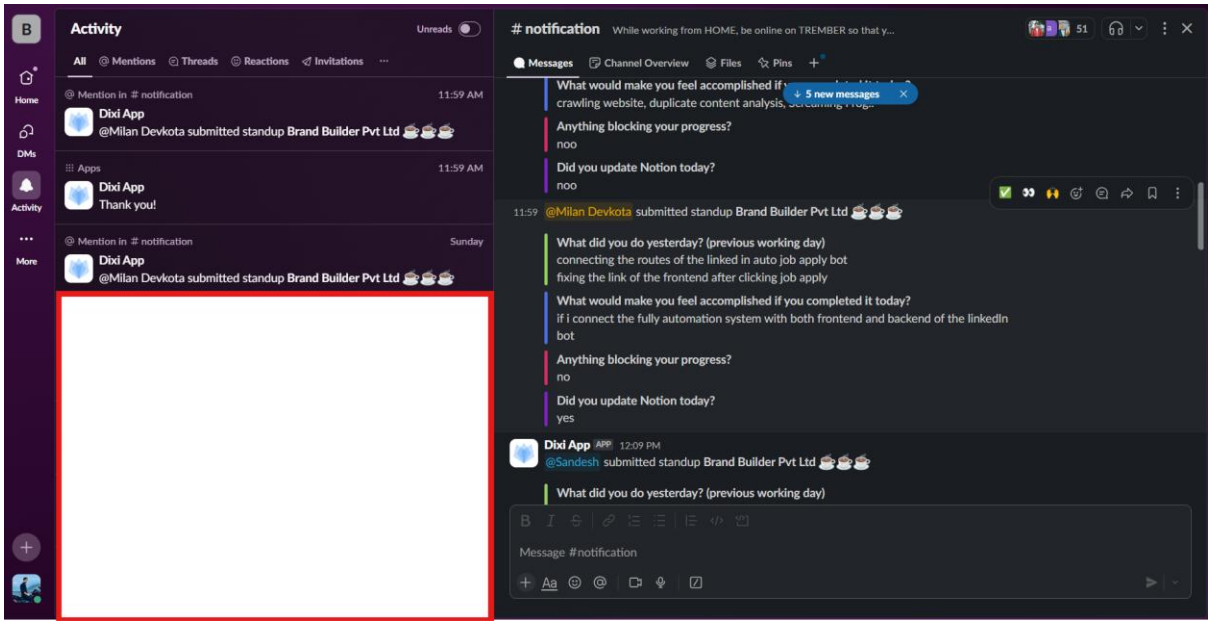

Fig: Slack for daily standups

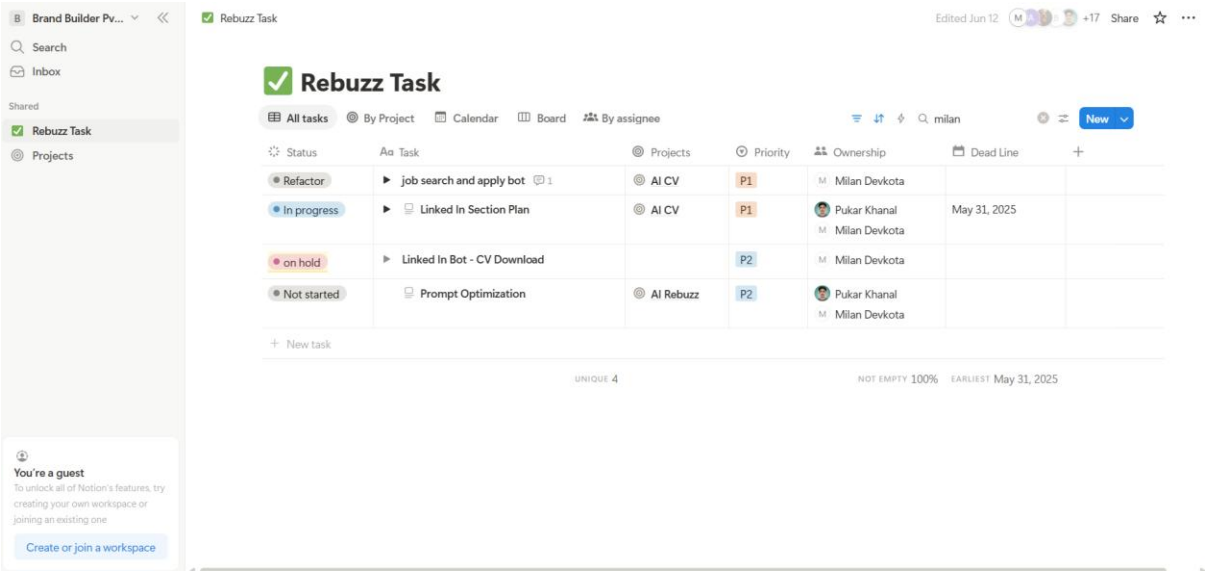

Fig: Notion for task monitoring

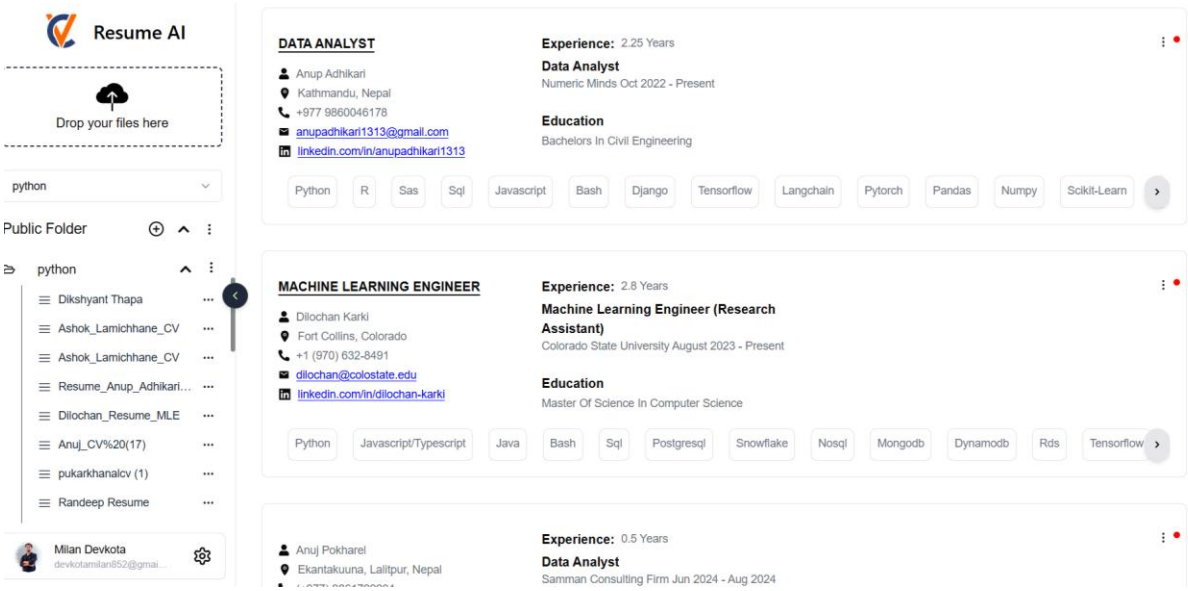

Fig: Resume AI dashboard

Mentor's Signed Logs of every week

Days	Activities (1st Week)	Signature
Day 1	Commenced internship with introduction and project overview. Set up development environment.	
Day 2	Conducted initial research on LinkedIn's website structure to identify key HTML elements for job searching and application automation.	
Day 3	Studied Selenium's documentation and tutorials to gain a comprehensive understanding of web automation techniques using Python.	
Day 4	Planned the project's architecture, defining the necessary modules for login, search, filtering, application processing, GUI, and backend API communication.	
Day 5	Established the project's repository using Git and created the initial code structure for the bot, GUI (HTML, CSS, JS), and backend (FastAPI).	
Day 6	Configured environment variables to securely handle sensitive information and set up Python virtual environments to manage project dependencies.	

Days	Activities (2nd Week)	Signature
Day 1	Developed a function using Selenium to automate the login process on LinkedIn, including handling username and password input.	
Day 2	Implemented cookie-based login functionality to persist user sessions and avoid repeated logins, enhancing efficiency.	
Day 3	Set up the basic structure for the GUI using HTML, focusing on the layout and initial elements.	
Day 4	Designed the GUI form to capture user input, including credentials, job search criteria, and resume data.	

Day 5	Established a basic FastAPI application and defined the initial API endpoints required for communication with the GUI.	
Day 6	Connected the GUI's login form to the FastAPI backend, enabling data transfer between the frontend and backend (without Selenium integration at this stage).	

Days	**Activities (3rd Week)**	**Signature**
Day 1	Developed the function to navigate to the LinkedIn Jobs section and initiate job searches using Selenium.	
Day 2	Implemented the job search functionality, enabling users to specify job titles and locations through keywords.	
Day 3	Added input fields to the GUI to allow users to enter their desired job title and location for the search.	
Day 4	Connected the GUI's job search inputs to the FastAPI backend, which triggered the Selenium job search process.	
Day 5	Implemented the display of job search results (initially job titles) within the GUI.	
Day 6	Implemented error handling for the job search process (Selenium and API) and displayed error messages in the GUI.	

Days	**Activities (4th Week)**	**Signature**
Day 1	Developed the function to filter job listings for "Easy Apply" jobs.	
Day 2	Integrated the "Easy Apply" filter option into the GUI.	
Day 3	Implemented file upload functionality in the GUI to allow users to upload their resume files.	

Day 4	Configured the FastAPI backend to receive and temporarily store the uploaded resume file.	
Day 5	Designed the data structure (JSON format) to store and manage resume information within the backend.	
Day 6	Implemented the function to load resume data from the uploaded file and store it in the backend.	

Days	**Activities (5th Week)**	**Signature**
Day 1	Developed the core logic to iterate through job listings and initiate the application process using Selenium.	
Day 2	Analyzed LinkedIn's Easy Apply forms to identify common form fields and their HTML elements.	
Day 3	Implemented functions to automatically fill basic application form fields using data from the user's resume.	
Day 4	Created API endpoints in FastAPI to trigger the application process for a selected job listing.	
Day 5	Connected the GUI to the FastAPI application endpoints to initiate job applications.	
Day 6	Conducted initial testing of the application automation for basic fields using sample Easy Apply job postings.	

Days	**Activities (6th Week)**	**Signature**
Day 1	Implemented functionality to handle resume uploads within the application process using Selenium.	
Day 2	Addressed scenarios with conditional form fields or variations in application structures, implementing initial handling strategies.	

Day 3	Developed a mechanism within FastAPI to track the status of each application and log attempts.	
Day 4	Incorporated delays and waiting times into the bot's actions to mimic human behavior.	
Day 5	Refined the GUI by adding progress indicators and detailed status messages.	
Day 6	Improved the CSS styling of the GUI for better user experience.	

Days	**Activities (7th Week)**	**Signature**
Day 1	Integrated LangChain with the FastAPI backend to handle custom application questions.	
Day 2	Set up the LLM (Gemini) and defined prompts to generate answers based on resume data.	
Day 3	Implemented logic to identify and extract custom questions from application forms using Selenium and FastAPI.	
Day 4	Developed the function to generate answers using LangChain and automatically fill them into the form.	
Day 5	Began exploring basic AI/ML concepts: supervised vs. unsupervised learning.	
Day 6	Gained a basic understanding of Natural Language Processing (NLP): text processing, tokenization.	

Days	**Activities (8th Week)**	**Signature**
Day 1	Implemented robust error handling to manage exceptions during the application process.	

Day 2	Added comprehensive logging to record detailed information about the bot's actions and errors.	
Day 3	Refined the bot's behavior to handle edge cases and unexpected scenarios.	
Day 4	Improved the bot's overall efficiency and speed by optimizing code and minimizing waits.	
Day 5	Enhanced the bot's ability to adapt to potential changes in LinkedIn's user interface.	
Day 6	Conducted thorough testing of the entire system (GUI, API, Selenium automation).	

Days	**Activities(9th Week)**	**Signature**
Day 1	Implemented advanced GUI features: progress bars, detailed logs, user settings.	
Day 2	Added basic user authentication to the GUI and FastAPI backend.	
Day 3	Optimized the GUI for responsiveness across various screen sizes.	
Day 4	Prepared the FastAPI application for deployment, exploring Docker.	
Day 5	Documented the API endpoints and GUI functionality.	
Day 6	Created a detailed deployment plan and checklist.	

Days	**Activities (10th Week)**	**Signature**
Day 1	Implemented enhanced handling for conditional logic in application forms.	
Day 2	Developed a module to extract and save application confirmation details after submission.	

Day 3	Integrated a mechanism to handle applications requiring external website redirects.	
Day 4	Optimized the bot's performance by implementing asynchronous task execution for certain operations.	
Day 5	Improved the bot's resilience to website changes by using more robust element selection strategies.	
Day 6	Conducted focused testing on the application handling module to identify and resolve any edge cases.	

Days	**Activities (11th Week)**	**Signature**
Day 1	Enhanced the GUI to display detailed application status updates to the user.	
Day 2	Implemented user settings within the GUI to allow customization of bot behavior.	
Day 3	Refactored the FastAPI backend to improve code organization and maintainability.	
Day 4	Integrated a rate-limiting mechanism into the backend to prevent excessive requests to LinkedIn.	
Day 5	Implemented logging within the GUI to provide users with insights into the bot's actions.	
Day 6	Conducted integration testing of the GUI and backend to ensure seamless communication and data flow.	